SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE CODE: CSA1514

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation
(Level 1) Question Count: 5

Total Marks: 25

Code of Conduct

I P .Gowtham Reddy(192525139)(Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student: P .Gowtham Reddy

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Total Marks:

1) Define cloud computing and explain any four characteristics that make it diffent from traditional, and premise computing.

A) Cloud computing is the delivery of computing resources such as server, storage, networking, database, and software over the internet. Instead of purchasing and maintaining physical infrastructure, user can access resources from cloud providers whenever required.

Four characteristics:

1. on-demand self services: user can create or access computing resources whenever required without direct interaction with the service provider.

2. Broad network access:

cloud services can be accessed through the internet using devices such as computers, tablets, and smartphones.

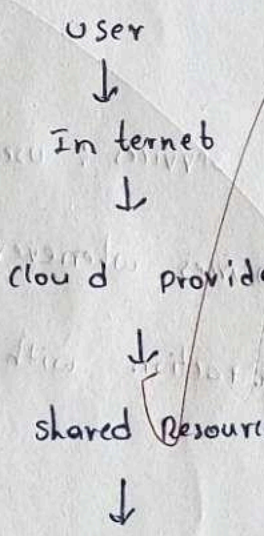
3. Resource pooling:

the provider shares a common pool of computing resource among multiple users using virtualization.

4. Rapid elasticity:

Resources can be increased or decreased quickly according to demand.

Flowchart



2) Trace the evalution of cloud computing from hardwave evolution to internet software evolution, and show how these changes enabled modern cloud platform.

A) Cloud computing developed throught several stages of hardware and software evolution.

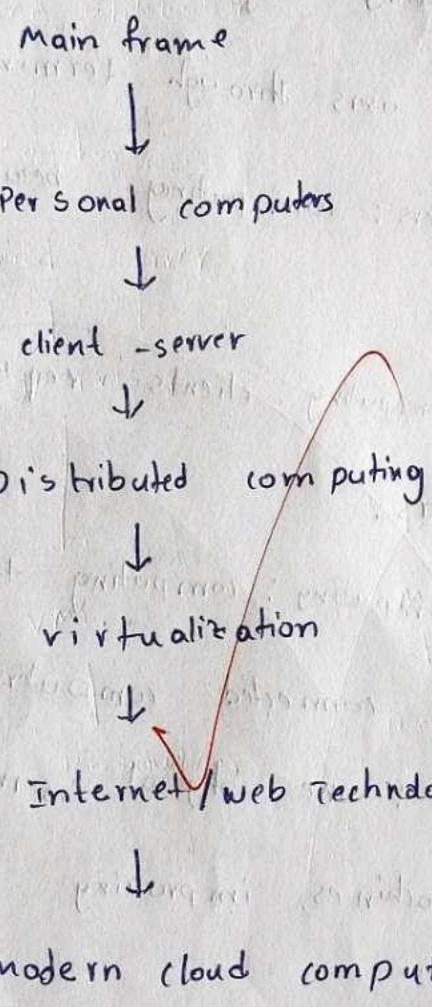
Evolution:

1. Mainframe computing: large centralized computers were shared by many users through terminals.
2. Personal computers: computing because available to individual users.
3. Client - server computing: clients requesting services from centralized servers.
4. Distributed computing: computing tasks were distributed acoross multiple connected computers.
5. virtualization: one physical server could run multiple virtual machines, improving resource utilization virtualization.

6. Internet and web evolution: High-speed internet and web technologies made remote access to computing resources possible.

7. Cloud computing: providers began offering computing resources as scalable services over than Internet.

Flow chart:



3) Explain the role of server virtualization in cloud computing. Differentiate between virtulization as an enabling technology and cloud computing as a service model.

A) server virtualization is a technolgy that divides one physical server into multiple idependent virtual machines (VMs) using a software layer called a hypervisor.

Role in cloud computing :

→ Improves utilization of physical server.

→ Allows multiple operating system to run on one physical machine.

→ Reduces hardware and maintenance costs.

→ Provides isolating between users.

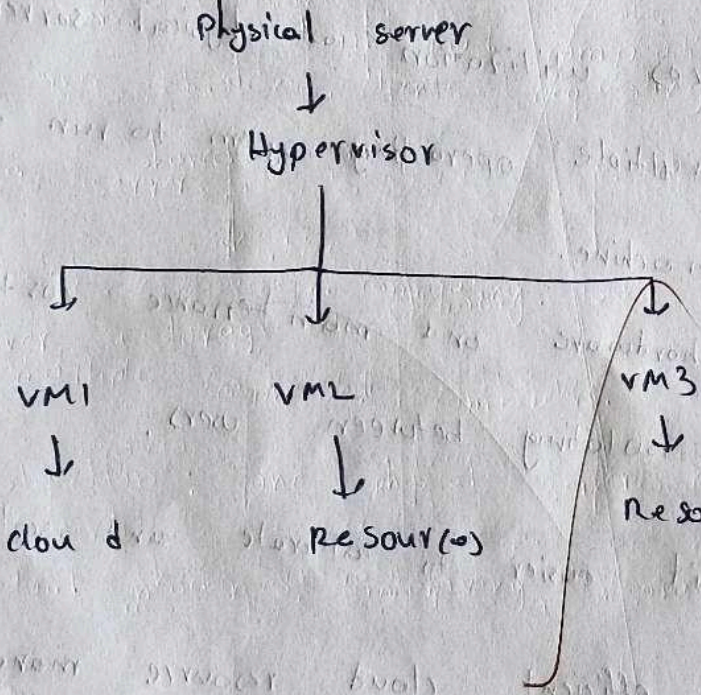
→ Makes it easier to allocate and scale resources

→ Supports efficient cloud resource management.

Virtulization vs cloud computing :

virtualization	cloud computing
→ It is a enabling technology	→ It is a service delivery model
→ creates virtual resources from physical resources	→ Delivers computing resources throught a network.
→ mainly focuses on resources abstraction and utilization	→ focuses on providing resources as services.

Flowchart :



4) Discuss the role of data center in delivering cloud services.

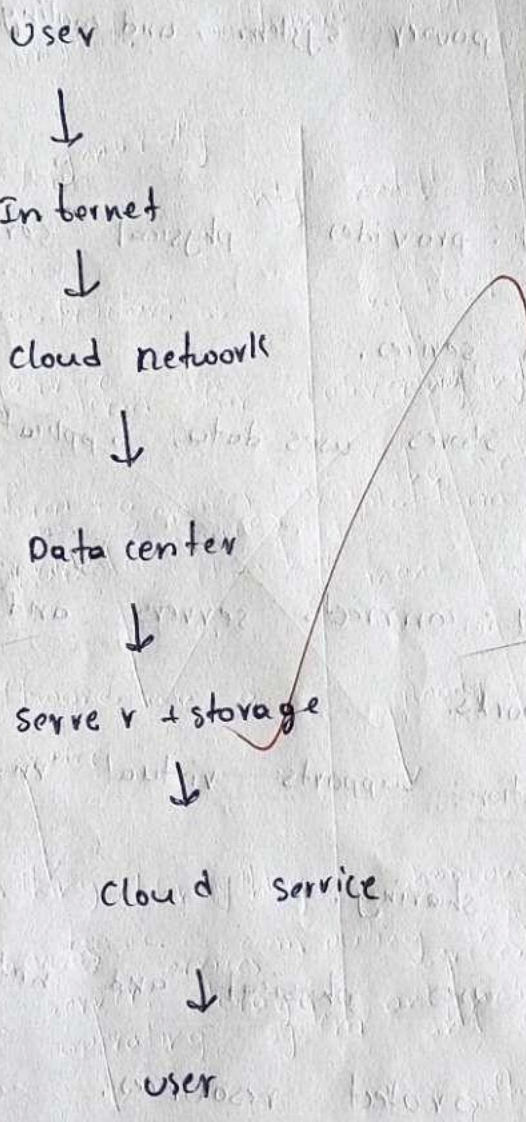
A) A data center is a facility containing computing infrastructure such as server, storage system, networking equipment, power system, and cooling system.

Major roles:

1. Computing: provides physical server to run application and cloud serices.
2. Storage: Stores uses data, application, data bases, and backs.
3. Networking: connects servers and users through high-speed networks.
4. Virtualization: supports virtual machine and efficient resource sharing.
5. Security: uses physical and network security mechanisms to protect resources.
6. Availability: uses backup power, redundant system, and multiples server to maintain service availability.

7. **Scalability** : Allows cloud provides to add more hardware when demand increases.

Flowchart:



5) Differentiate Iaas, Paas, Saas, and Xaas with one suitable real-world example for each model.

A) Cloud services can be provided at different levels depending on how much the cloud provider manages.

Model	Full form	what it Provides
Iaas	Infrastructure as a service	virtual machines, storage, networking
Paas	platform aa a service	platform, runtime, development tools.
Saas	softwares aa a service	Ready-to-use software.
Xaas	Anything / Everything aa a services	various resources or services delivered through the cloud.

1. Iaas :-

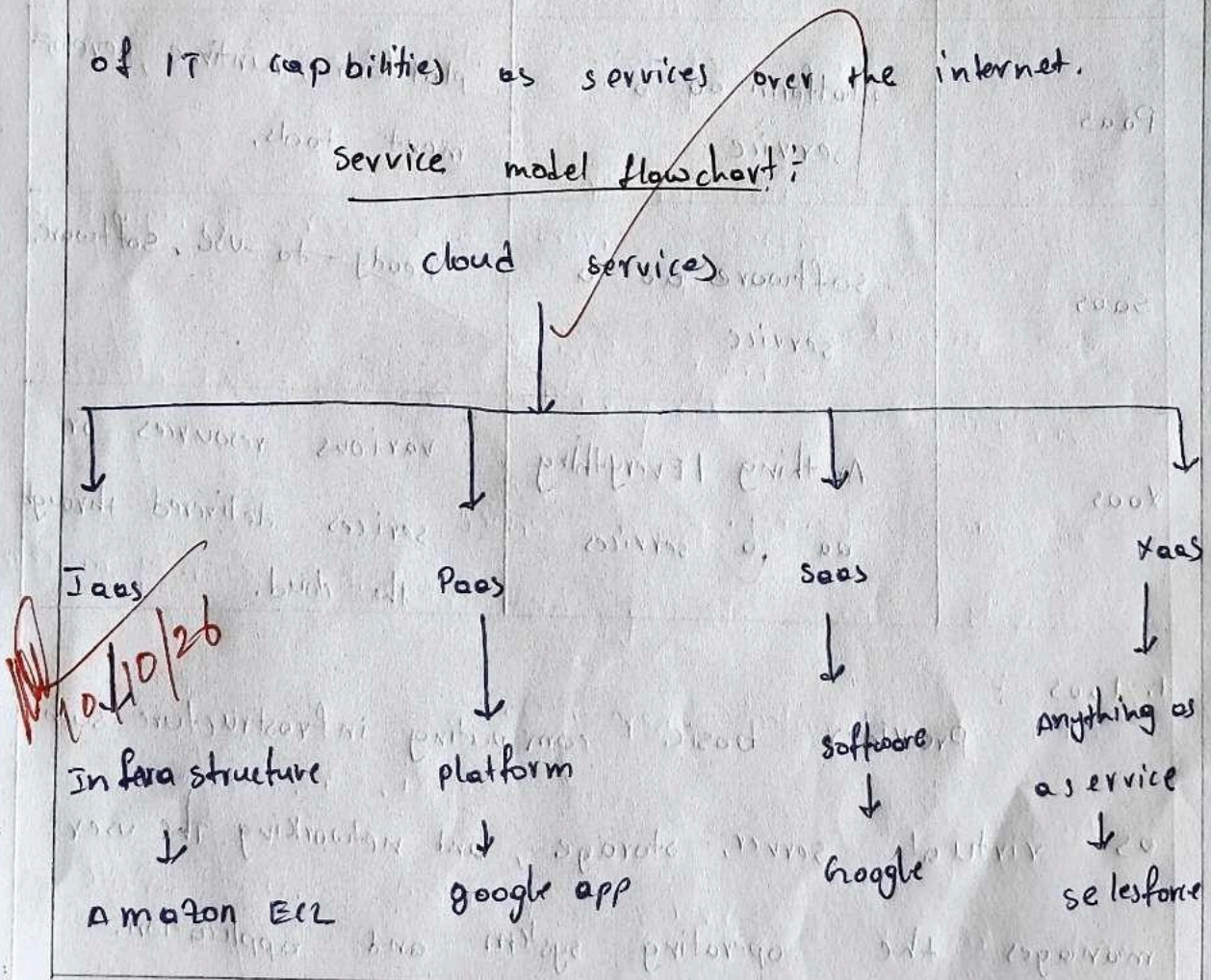
Provides basic computing infrastructure such as virtual server, storage, and networking. The user manages the operating system and application.

2. Paas :- Provides a redy-made platform for developing and deploying applications. Developers can focus mainly on application development instead of managing infrastructure.

3. Saas :- Provides complete software applications throgh the internet. users generally only need to use the application.

4. Xaas :- A broad term for delivering different types of IT capibilities as services over the internet.

Service model flowchart :-



10/10/26